L03_Williane_Yarro_ITAI3377

Deploying a Simple AI Model on a Simulated Edge Device

Course: ITAI-3377
Student: Williane Yarro

Instructor Name: Sitaram Ayyagari

Objective

The aim of this lab was to implement a basic artificial intelligence model on a simulated edge device to grasp the core principles of edge computing and the integration of AI.

Environment Setup

The laboratory setup utilized Python, Visual Studio Code, TensorFlow, and TensorFlow Lite. A virtual environment was established to guarantee compatibility with TensorFlow and to efficiently manage dependencies. Visual Studio Code served as the main development environment for writing and running the Python scripts on the Terminal

---

Dataset Preparation

The MNIST dataset served as the foundation for this lab. It comprises grayscale images depicting handwritten digits. The images underwent normalization by scaling pixel values to a range between 0 and 1, and were reshaped to satisfy the input specifications of a convolutional neural network.

Model Training

A convolutional neural network (CNN) was developed using TensorFlow and Keras. The model included convolutional, pooling, and dense layers and was trained for multiple epochs. The trained model achieved high accuracy when evaluated using the test dataset.

Edge Deployment (Simulation)

Following the training phase, the model was transformed into TensorFlow Lite format to emulate deployment on an edge device. TensorFlow Lite facilitated local inference on the CPU. This procedure mimicked edge computing by executing predictions directly on the device, eliminating the need for cloud services.

Testing and Validation Results

The model underwent testing with a sample consisting of 50 images. The results obtained during the simulated edge deployment were as follows:

Accuracy: 96.00%

Average inference latency: 0.06 milliseconds

Model size: 683.99 kilobytes

These findings indicate that the model operates effectively in an edge computing setting, exhibiting low latency and a compact memory footprint.

Conclusion

This laboratory effectively showcased the deployment of a basic AI model on a simulated edge device. The use of TensorFlow Lite facilitated rapid, low-latency inference and optimized resource utilization, emphasizing the benefits of edge computing for real-time applications.